Founder Memo — COD RTO Decision Engine

Objective: reduce avoidable COD RTO cost without sacrificing high-value first-time customer growth.

What the data says

The supplied dataset contains 2,018 COD orders with an observed RTO rate of 28.2%. Three findings stood out. First-time customers had a 30.2% RTO rate, versus 15.4% for customers with exactly one prior order. That makes customer history useful, but it is not a reason to cancel every first order. Second, acquisition source matters: Meta Ads had a 31.2% RTO rate versus 22.7% for direct traffic. Third, courier performance differs: Xpressbees was at 31.2% across 397 orders. A useful warning is that small groups can look extreme: Gorakhpur showed 63.0% RTO, but only across 46 orders, so I would not create a city-wide block from that signal alone.

Recommended policy

I recommend a four-bucket policy: below 20% predicted RTO, ship as-is; 20–35%, request WhatsApp confirmation; 35–60%, offer ■30 off for prepaid; above 60%, hold/cancel. A suspicious placeholder name can strengthen a cancellation decision when risk is already high. These thresholds live in config.yaml so Operations can change them without engineering work.

The model uses numerical history for the risk score. An LLM is only used where language is genuinely messy: checking names such as 'asdf' and interpreting incomplete or landmark-based Hinglish addresses. This keeps the system predictable and inexpensive.

Operations vs Marketing in ■

The stated RTO cost is ■65 forward shipping + ■65 return shipping + ■25 handling + 8% × ■180 damage exposure = ■169.40 per RTO. The stated prepaid RTO rate is 4% versus 28% for COD, a 24-point difference. At the stated rates, that gap is worth about ■40.66 of expected RTO cost per order before payment/discount costs. A ■30 prepaid incentive therefore has a plausible positive unit-economic case, but the exact conversion effect must be tested.

The marketing concern is equally important. If a first-time Meta customer is cancelled, the company can lose the gross margin from a genuine order plus roughly ■300 acquisition cost. The policy therefore uses confirmation and prepaid nudges as intermediate actions rather than treating risk as a binary cancel decision.

What last month would have looked like

Using September 2026 (the last month in the supplied dataset), the model routed 334 orders as follows: 1 ship-as-is, 38 confirmation, 233 prepaid, and 62 hold/cancel. Under conservative planning assumptions (50% RTO reduction for confirmation and 80% for prepaid), the estimated RTO-cost avoided is ■12,569. However, the same historical simulation shows an estimated ■24,782 of gross-margin/CAC value among cancelled orders that actually delivered. With the conservative assumptions, the modeled net impact is ■-17,733, which is negative. That is an important result: the initial thresholds are too aggressive and should be relaxed before production.

This is exactly why I would not ship the first version with automatic cancellation at 60% as a hard business truth. The prototype exposes the thresholds so the team can tune them and should run an A/B test on confirmation/prepaid conversion before expanding cancellation.

What could go wrong

The model can overfit sparse city/pincode history, addresses can be valid but look unusual, customers may ignore WhatsApp, and a courier change can make historical rates stale. I would monitor RTO rate by action bucket, cancellation-to-delivered rate, prepaid conversion, confirmation rate, false-positive rate, and contribution margin. Alerts should fire if cancellation of subsequently-delivered orders rises or if RTO improves in one bucket but margin falls.

Next two weeks

I would add time-aware retraining, stronger pincode/phone-history features, an actual WhatsApp provider adapter, an experiment framework for ■20/■30/■40 prepaid incentives, calibrated probability estimates, and a weekly threshold-optimization report based on contribution margin rather than accuracy alone.